VoltGuard Rwanda: Electricity Billing and Meter Tamper Monitor

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Problem Statement

A utility must apply the correct tariff to every meter reading, and tariffs change: the Rwanda Utilities Regulatory Authority (RURA, 2025) revised the end-user tariff with effect from 1 October 2025. Faulty or manipulated readings and billing errors are a recognised source of lost revenue, called non-technical losses (Glauner et al., 2016). Without rules in the database, a reading lower than the previous one, or a bill calculated with an outdated tariff, can pass unnoticed. Public information shows how customers buy prepaid electricity (REG, n.d.) but not how readings are validated and audited, which is the part this project models.

Proposed Solution

VoltGuard is an Oracle database with PL/SQL that stores meters, readings, versioned tariffs and bills. The main objective is to bill each reading with the correct tariff and to reject and log suspicious readings. Band limits change too: the first residential band was below 15 kWh in 2020 (RURA, 2020) and is 0-20 kWh from October 2025, so tariffs must be versioned. Specific objectives: (1) normalise real consumption data and the RURA tariffs to 3NF; (2) bill each reading with the tariff version valid on its date; (3) reject readings lower than the previous one; (4) audit changes to readings, bills and tariffs; (5) give staff and customers a simple app.

Data Source

UCI Machine Learning Repository, Individual Household Electric Power Consumption (Hebrail & Berard, 2012; CC BY 4.0): 2,075,259 minute readings, 9 attributes, December 2006 to November 2010, one anonymous household in France. The RURA tariff documents (2020 and 2025) are the second real source. Because the household is French, customers, meters and districts are simulated, and consumption is scaled to Rwandan averages of about 22 kWh per month (Mugyenyi et al., n.d.). All simulated data is labelled as such.

Planned Tables

DISTRICT, CUSTOMER_CATEGORY, CUSTOMER, METER, METER_READING, TARIFF_VERSION, TARIFF_BAND, BILL, TAMPER_EVENT, AUDIT_LOG, and a staging table STG_UCI_POWER for the raw file, all in 3NF with primary and foreign keys. A customer has many meters, a meter has many readings, a tariff version has many bands, and each bill points to one tariff version.

Planned PL/SQL Features

A trigger on METER_READING rejects a reading lower than the previous one, and an autonomous transaction still saves the attempt in TAMPER_EVENT. The function fn_energy_charge bills a reading by the tariff version valid on its date. Procedures record_reading and generate_bill are the only way to add readings and create bills. A cursor reports monthly consumption per district. Audit triggers write to AUDIT_LOG, a restriction rule limits tariff changes, custom exceptions cover unknown meters and missing tariffs, and the package volt_pkg groups the code.

Innovation

Option: User interface. Tool: Oracle APEX. Staff enter readings and generate bills, each action calling a PL/SQL procedure, and customers look up a bill through an Oracle REST Data Services endpoint. The rules stay in PL/SQL, and the app will be demonstrated live on the real Oracle data.

References (APA)

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